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24EC2210 Home Assignment - 2 2400040454

1. Consider the message sender wants to send 110100101, and the generator Polynomial is $x^5+x^4+x^2+1$. Find the message transmitted by the sender. if the reciever recieves the message. check if the reciever recieve Correct message or not?

Sol:

Given Data:

Message: 110100101

Generator polynomial: $x^5+x^4+x^2+1$

Compare G(p): $1.x^5+1.x^4+0.x^3+1.x^2+0x+1$

redundant bits: 110101 = 5 add 5 zero to message

∴ perform XOR operation

```
          10000100
110101 | 11010010100000
         110101
Discard ← 0000011
          000000
Discard ← 0000110
           000000
Discard ← 0001101
            000000
Discard ← 0011010
             000000
Discard ← 0110100
              110101
Discard ← 0000010
               000000
Discard ← 0000100
                000000
Discard ← 0001000
                 000000
Discard ← 001000
```

Replace appended 5 bit zero's With CRC : 01000

∴ Now Verification from Reciever Side

Reciever Side :→

```
              10000100
      110101 ) 110100101010000
               110101↓
   Discard ← 0 00001 1
               000000
   Discard ← 0 000110
               000000
   Discard ← 0 001101
               000000
   Discard ← 0 011010
               000000
   Discard ← 0 110101
               110101
   Discard ← 0 000000
               000000
   Discard ← 0 000000
             ← 0 000000
   Discard ← 0 000000
               000000
               00000
```

∴ Since the remainder obtained at the reciever is zero, the recieved message error free.

2) Identify the total delay (latency) for a frame of size 5 million bits that is being sent on a link with 10 routers, each having a queueing time of 2μsec and processing time of 1μsec & the link has bandwidth of 5Mbps. Which component of the total delay is dominant ? which one is negligible ? Assume the length of link 2000km. the speed of light inside link is 2×10^8 m/s ?

Sol: Given Data :

Frame size (L) = 5 million bits = 5×10^6 bits

Bandwidth (R) = 5Mbps = 5×10^6 bps

Number of routers = 10

Queuing time per router = 2μs

processing time per router = 1μs

link length = 2×10^6 m

propagation speed = 2×10^8 m/s.

Total Delay = Transmission Delay + Propagation Delay + Processing Delay + Queuing Delay.

1. Transmission Delay $= \frac{\text{Frame Size}}{\text{Bandwidth}} = \frac{5\times10^6}{5\times10^6} = 1$

2. Propagation Delay $= \frac{\text{Link length}}{\text{propagation speed}} = \frac{2\times10^6}{2\times10^8} = 0.01s = 10ms$

3. Processing Delay $= 10\times10^6 = 10^6 s$

4. Queuing Delay $= 10\times2\times10^6 = 20\mu s$

Total Delay $= 1 + 2 + 3 + 4$

$= 1 + 0.01 + 0.00001 + 0.00002$

$\boxed{\text{Latency} = 1.01003 \text{ seconds}}$

- Dominant delay Component : Transmission delay (1 second).
- Negligible delay Component : processing delay and queuing delay.

3. A reciever recieved a 14 bit Hamming Code as 11011011110101, assuming even parity and stated wheather the recieved data is Correct or not, if not locate error bit?

Sol: Recieved Hamming Code : 11011011110101 = 14 bits

Parity used : Even parity

We know that in even parity, the number of 1's must be even

Parity position are must be 2^n = 1, 2, 4, 8

1, check P_1 position = 1, 3, 5, 7, 9, 11, 13

Bits at position = 1, 0, 1, 1, 1, 1, 1

sum of 1's = 6 (even)

$s_1 = 0 \rightarrow$ 1

2, check parity P_2 : 2, 3, 6, 7, 10, 11, 14

Bits at position : 1, 0, 0, 1, 1, 1, 1

sum of 1's = 5(odd)

$S_2 = 1 \longrightarrow$ (2)

3, Check parity P_4 : 4, 5, 6, 7, 12, 13, 14

Bits at position : 1, 1, 0, 1, 0, 1, 1

sum of 1's = 5(odd)

$S_4 = 1 \longrightarrow$ (3)

4, check parity P_8 : 8, 9, 10, 11, 12, 13, 14

Bits at position : 1, 1, 1, 1, 0, 1, 1

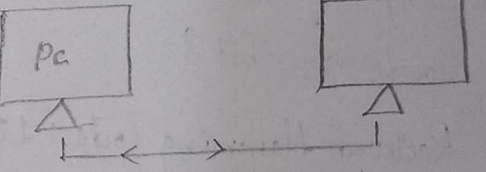
sum of 1's = 6(even)

$S_8 = 0 \rightarrow$ (4)

Form the Syndrom : $0110_2 = 6_{10}$

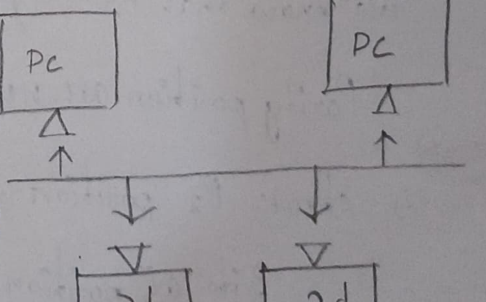
4. Design and analyze all the networks topologies, by providing detailed explanation and neat sketch for each, and demonstrate how they impact network perfomance and Scalability?

1) Point - to - point :→



Two device are directly Connected with one link. it gives fast Communication and low delay, but it Cannot support many device. it simple and reliable, but not Scalable.

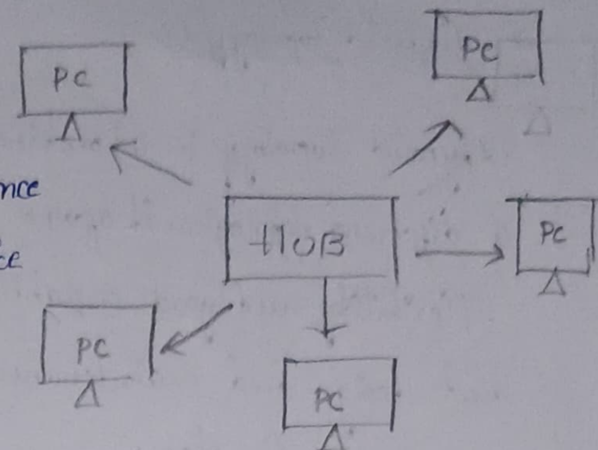
2) Bus Topology :→



All the devices share a Single cable to Communicate performance reduce when many devices are Connected because bandwidth is shared. it is cheap and easy to install, but failure of the main cable stops the entire network.

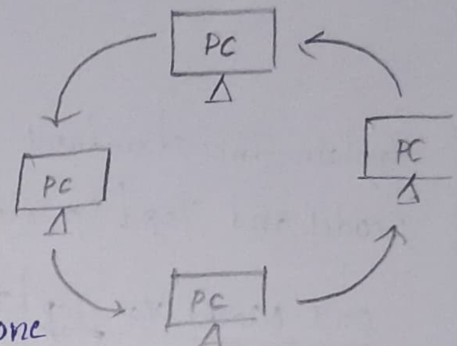
3. Star Topology :→

All the devices are Connected to a Central hub or Switch. it provides good perfomance and easy to expand, but if the central device fails, the whole network goes down.



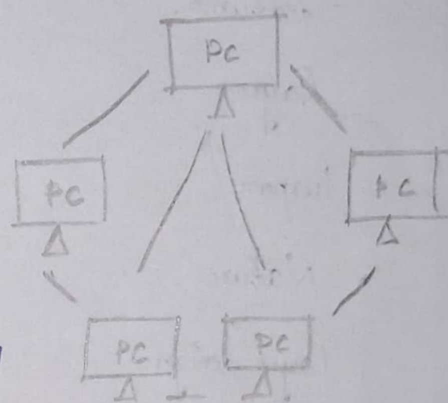
4. Ring Topology :→

Devices are Connected in a Circular path. Dataflow in one direction, avoiding Collision. but adding more device increase delay. failure of one device or Cable Can affect the entire network.



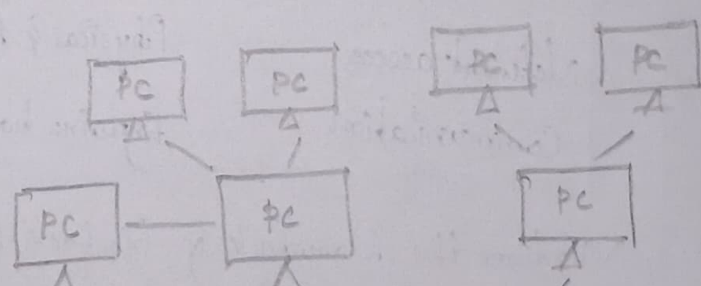
5) Mesh Topology :

Each device is Connected to many other device. it provide high reliability and perfomance, but it is expensive and difficult to manage. Scalability is poor in full mesh network.



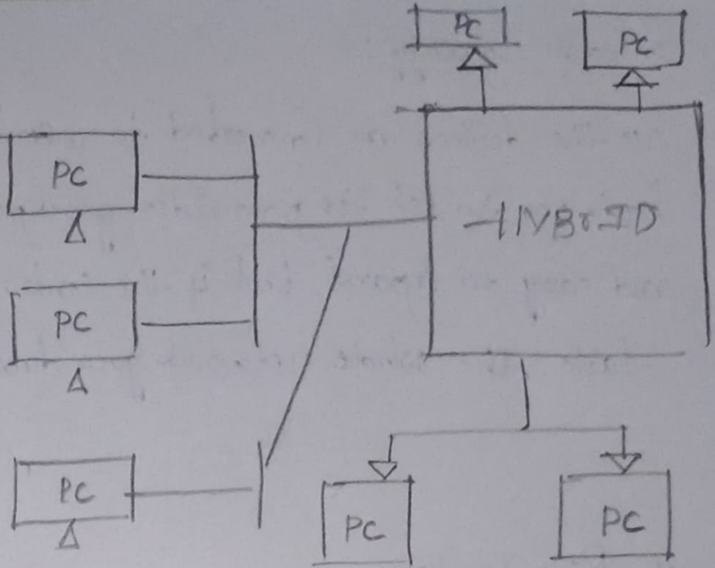
6) Tree Topology :

Device are Connected in a heirarchical structure. it supports easy expansion and good perfomance, but failure of high level affect many nodes.



7. Hybrid Topology :→

Hybrid Topology is a Combination of different topologies. it offers flexibility and good scalability but design and maintenance are more Complex.



5, Explain the structural and functional difference between the OSI reference model and Tcp/Ip model used in Computer network ?

Ans: OSI Model Vs Tcp/Ip model :

Features	OSI Model	Tcp/Ip model
Layer	7 layer	4 layers
Purpose	learning & reference	real network
Nature	Theoretical model	practical model
Layer design	Functions are superate	function are Combined
session & presen	separate layer	Part of application layer
Network access	Physical & Data superate	Physical & Data Combined
Communication	Explains how data should	Explain how data actually

6. Mention the Drawback of Go-back-N sliding window protocol and discuss how it Can be prevented by the selective Repeat protocol ?

Ans:

- If one frame is lost or damaged, all following frames are discarded.
- Sender retransmit many frame unnecessarily.
- Waste bandwidth.
- Reduce efficiency in noisy network.

- Reciever accept and stores Correct frame.
- only lost or damaged frames are transmitted.
- Reduce unnecassary transmission.
- improve bandwidth efficiency.

7, suppose the following block of 32 bits is to be sent using a checksum 8 bits 11100101, 01101011, 10101111, 01011011 and check your answer from reciever 10110101, 01100011, 10101111, 11011011 ?

sol: • Checksum size = 8 bit

Sender side: Calcalation

11100101
01101011
1 01010000
1
01010001
10101111
1 00000000
1
00000001
01011011
01011100

Data sum = 01011100
1's complement = 10100011

Reciever Side: Calculation

10110101
01100011
1 00011000
1
00011001
10101111
11001000
11011011
1 100100011
1
10010100
10100011
100110111
1
00111000

1st Complement = 11000111

∴ Reciever result not all zero so recieved data Contain errors.

8\. Discriminate between the Send window & recieve the window for links and how are they related to (i) a selective repeat transmission scheme 2) a Go back N control scheme?

Send Window:

- set of frames the Sender is allowed to send without waiting
- Controls how many frame Can be transmit
- Size depends on the flow error

Recieve Window:

- set of frames the receiver is ready to accept.
- Control which frame number are acceptable.
- Helps in handling out-of-order frames.

Select Repeat (SR)

- Can send multiple frames.
- Retransmit only the lost or damaged frames
- size is greater than 1.
- Reciever accept out of order and stores them.

Go-Back-N (GBN)

- Can send multiple frames.
- on error, retransmits the errored frames and all following
- size is 1.
- Reciever accept only the excepted frame.